DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

EVALUATION SCHEME & SYLLABUS

FOR

B. TECH. FOURTH YEAR

(BIOMEDICAL ENGINEERING)

On

Choice Based Credit System
(Effective from session 2019-20)

EVALUATION SCHEME

B.TECH. BIOMEDICAL ENGINEERING

YEAR 4th / SEMESTER VII **Effective from session 2019-20**

Sr. No.	Sub Code	Subject Name	Dept.	L-T-P	Th/Lab Marks	Sessional		Subject Total	Credit
					ESE	CT	TA		
1		Open Elective-I**	Other Dept.	3--0--0	70	20	10	100	3
2		Departmental Elective-III	Core Deptt.	3--0--0	70	20	10	100	3
3		Departmental Elective-IV	Core Deptt.	3--1--0	70	20	10	100	4
4	RBM701	Biomedical Image Processing	Core Deptt.	3--1--0	70	20	10	100	4
5	RBM702	Hospital Management System	Core Deptt.	3--0--0	70	20	10	100	3
6	RBM751	Biomedical Image Processing Lab	Core Deptt.	0--0--2	50	-	50	100	1
7	RBM752	Medical System Lab	Core Deptt.	0--0—2	50	-	50	100	1
8	REC753	Industrial Training Viva-Voce	Core Deptt.	0--0—3	-	-	100	100	2
9	REC754	Project-I	Core Deptt.	0--0--6	-	-	200	200	3
	TOTAL				450	100	450	1000	24

Departmental Elective – III

1. RBM071 Bioinformatics
2. RBM072 Artificial Organs & Rehabilitation Engineering
3. RBM073 Therapeutic Equipments

Departmental Elective – IV

4. RBM074 Telemedicine
5. RBM075 Biotransport Phenomenon
6. RBM076 Data Communication Networks

EVALUATION SCHEME

B.Tech. BIOMEDICAL ENGINEERING

YEAR 4th / SEMESTER VIII **Effective from the Session 2019-20**

Sr. No	Sub Code	Subject Name	Dept.	L-T-P	Th/LAB Marks	Sessional		Subject Total	Credit
					ESE	CT	TA		
1		Open Elective-II**	Other Dept.	3-0-0	70	20	10	100	3
2		Departmental Elective-V	Core Deptt.	3-1-0	70	20	10	100	4
3		Departmental Elective- VI	Core Deptt.	3-0-0	70	20	10	100	3
4	REC851	GD &Seminar	Core Deptt.	0-0-3			100	100	2
5	REC852	Project	Core Deptt.	0-0-12	350	-	250	600	12
	TOTAL				560	60	380	1000	24

Departmenal Elective – V

1. RBM081 Advanced Biomedical Instrumentation
2. RBM082 Tissue Engineering
3. RBM083 Principles of Radio Diagnosis and Radio Therapy

Departmental Elective- VI

4. RBM084 Design and Modelling of Biomedical Systems
5. RBM085 Communication Engineering
6. RBM086 Artificial Intelligence & its Applications in Biomedical Engineering

RBM701 BIOMEDICAL IMAGE PROCESSING

UNIT-I IMAGE PERCEPTION

Introduction, light, luminance, brightness and contrast, MTF of the visual system - visibility, function, monochrome vision models, color representation, color matching and reproduction, color vision model Image sampling and quantization, Image quantization, visual quantization.

UNIT-II IMAGE ENHANCEMENT & IMAGE RESTORATION

Point operations; contrast stretching, clipping and threshold, digital negative intensity level slicing, bit extraction, Histogram modeling, histogram equalization, modification, Convolution theorem and correlation, spatial operations, smoothing techniques.

UNIT-III IMAGE TRANSFORMS

Two dimensional orthogonal and unitary transforms, properties of unitary transforms — one dimensional DFT, cosine, sine Harmrd and Haar transforms.

UNIT-IV IMAGE SEGMENTATION ANLYSIS AND COMPUTER VISION

Spatial feature extraction, transforms features, Segmentation techniques, Analysis techniques.

UNIT-V

Application of MATLAB for Digital image processing.

BOOKS:

1. Jain Anil K., "Fundamentals of Digital Image Processing.", Prentice Hall.
2. Refael C. Gonzalez. Wintz Paul, "Digital Image Processing.", Addision Wesley.
3. Pratt W.K., "Digital Image Processing.", John Wiley and Sons.

RBM702 HOSPITAL MANAGEMENT SYSTEM

UNIT I

Classification of hospital & architecture: General hospital, specialized hospital, primary health care – their role and functions. Aspects of hospital services – inpatient, outpatient and emergency. Location and environment of hospital, Hierarchy of medical and paramedical staff & their functions and responsibilities. Modern Hospital Architecture- space in a hospital building, design of ward, intensive care units, air conditioning, plumbing & sanitation, gas supply, waste disposal, cleaning, dietary, sterilizing, laundry, storage and operation theatre systems, Radiology, Central labs, Blood banks, OPD, Casualty, etc.

UNIT II

Elements of Safety - Safety Publications and Standards Organizations - Orientation to Laboratory Safety - Types of risks in the hospitals - factors of environment - Safety showers and Eye Washes – Radiation hazards – radiation detection – safety measures – standards. Ergonomics - Flammables and Explosives – Formaldehydes - PEL Standards and Calculations - Material Safety - Organization of Safety in the hospitals. Electrical power systems in hospitals: Safety of electrical systems, Protective systems - interference of patient's protection grounding. Design of sub stations, breakers, Surge protectors, EMI filters, voltage stabilizers, generator sets and UPS. Uninterrupted power supply for ICU and computerized monitoring units. Specification & estimation for hospital wiring - small case study.

UNIT III

Air conditioning & gas supply systems: Air conditioning and refrigeration systems for small and large areas. Air changes, filtering and sterility. Deodourization, disinfection, dehumidification and cryogenic systems. Centralized supply of air, oxygen, nitrous oxide & vacuum - Principle of production of liquid oxygen. Management lifts fire fighting equipments.

UNIT IV

Hospital engineering & Management: Definition of biomedical Engineering, clinical engineering & hospital engineering. Importance of BME department – servicing and maintenance, testing, acceptance & maintenance protocols, Computerized preventive maintenance planning, MROs. Training of men for medical equipments preventive and periodical maintenance procedures. Preparation of estimates, specifications, tender details etc. Importance of ISO 9000 Certificates - Obtaining ISO certificates in hospitals. Proposed protocols. Necessity for standardization, FDA, AERB, Joint Commission of Accreditation of hospitals, ICRP and other standard organization, methods to monitor the standards.

UNIT V

Hospital Information system: Role of database in HIS. Need of Networking in HIS. Overview of Networking, topologies and its configuration. Structuring medical records to carry out functions like admissions, discharges, treatment history etc. Computerization in pharmacy & billing. Automated clinical laboratory systems & radiology information system. Need for evolving health policy, health organization in state, health financing system, health education, health insurance, health legislation.

TEXT BOOKS

1. P.E.Stanley, Handbook of hospital safety, CRC Press (UNIT II)
2. Arun Kumar, Hospital Management, Anmol Publications Pvt. Ltd., Jan 2000 , 1st.ed (UNITS IV & V)
3.Harold E. Smalley, "Hospital Management Engineering – A guide to the improvement of hospital management system", PHI.

REFERENCES:

1. Sharma, Essentials for Hospital Support Services and Physical Infrastructure, 1/e, Jaypee Medical Publishers 2003
2. Hospital Engineering And Facilities Management 2007 - Report, Fifth official report of the International Federation of Hospital Engineering (IFHE), January 2007
3. Gupta, Kant, Chandrashekhar, Satpathy, Modern Trends in Planning and Designing of Hospitals Principles and Practice with CD-ROM, Jaypee Medical publishers, 1/e, 2007
4. Sakharkar, Principles of Hospital Administration and Planning, Jaypee Medical publishers 1/e, Reprint 2004

RBM751 BIOMEDICAL IMAGE PROCESSING LAB

1. Study of MRI Images.
2. Study of CT Scan.
3. Study of Mammograms.
4. Reconstruction of Images.
5. Image analysis.
6. MATLAB implementation.

REFERENCE BOOKS:

1. Pathology & Micro Biology Laboratory Manual

RBM752 MEDICAL SYSTEM LAB

1. pH meter : Study, standardisation & calibration.
2. Calorimeter
3. Spectro Photometer
4. Flame photometer
5. Hb meter
6. Conductivity meter
7. Study & familiarization of Laser Equipments
8. Study of physiological pre-amplifiers.
9. pressure measurements using physiological transducers.
10. Servicing of ECG equipments.
11. Study of multi channel physiological recorders.
12. Study of vacuum tube and solid state cautery.
13. Study of ventiltor.
14. Study of ultrasonic equipment.
15. Study of X-ray radiography system.

DEPARTMENT ELECTIVE –III

RBM071 Bioinformatics

UNIT-I INTRODUCTION TO BIOINFORMATICS

Objectives of Bioinformatics, Data integration, Data Analysis, Bioinformatics databases and tools, Molecular approach versus Bioinformatics approach, Overview of Bioinformatics application.

UNIT-II MOLECULAR BIOLOGY AND INFORMATION

Basic chemistry of nucleic acids, Structure of DNA, Genes, The functional elements in DNA, DNA sequencing and Polymeric chain reaction, Cloning methodology, Amino acids, Protein structure Protein folding, Protein function.

UNIT-III SEQUENCE ALIGNMENT

Introduction to Sequence Analysis, Models for sequence analysis and their Biological motivation, Methods of alignment, Usage of gap penalties and Scoring matrices, Tools for sequence alignment, Tools for multiple sequence alignment, Applications of Multiple alignment.

UNIT-IV GENE MAPPING AND GENE EXPRESSION

Applications of Gene mapping, DNA sequencing, DNA micro arrays, Algorithms for gene alignment, Gene prediction tools, Tools for DNA/RNA structure and function analysis.

UNIT-V PROTEOMICS

Protein structure visualization, Protein structure prediction, Methods of protein structure for known folds, Methods of protein structure for unknown folds, Methods for structure prediction, Protein analysis, Tools for protein analysis.

BOOKS

- Rastogi S.C., Namita Mendiratta, Parag Rastogi, "Bioinformatics Concepts, Skills and Applications", CBS publication.
- Baxevanis A.D, Francis Ouellette, "Bioinformatics: A practical guide to the Analysis of Genes and Proteins", Wiley Interscience, New York.
- Mount David, "Bioinformatics Sequence and Genome Analysis", Cold Spring Harbor Laboratory Press.
- Hooman Rashidi, and Lukas K.Buehler, "Bioinformatics Basics Applications in Biological Science and Medicine", CRC Press.
- Tisdall James, "Beginning Perl for Bioinformatics", O'Reilly publications.

RBM072 ARTIFICIAL ORGANS & REHABILITATION ENGINEERING

UNIT I

Introduction to artificial organs: Biomaterials used in artificial organs and prostheses, inflammation, rejection, correction. Rheological properties of blood, blood viscosity variation: effect of shear rate, hematocrit, temperature and protein contents. Casson equation, flow properties of blood through the blood vessels, problems associated with extracorporeal blood flow.

UNIT II

Artificial kidney: Brief of kidney filtration, basic methods of artificial waste removal, hemodialysis, equation for artificial kidney and middle molecule hypothesis. Hemodialysers: flat plate type, coil type and hollow fiber. Analysis of mass transfer in dialyers (cross current & cocurrent flow), regeneration of dialysate, membrane configuration, wearable artificial kidney machine, separation of antigens from blood in ESRD patients.

UNIT III

Artificial heart-lung machine: Brief of lungs gaseous exchange / transport, artificial heart-lung devices. Oxygenators: bubble, film oxygenators and membrane oxygenators. Gas flow rate and area for membrane oxygenators. Liver support system, artificial pancreas, blood and skin.

UNIT IV

Audiometry: air conduction, bone conduction, masking, functional diagram of an audiometer. Hearing aids: different types, receiver amplifiers. Opthalmoscope, retinoscope, I.A.B.P principle and application.

UNIT V

Rehabilitation Engineering: Impairments, disabilities and handicaps, Measurement and assessment. Characterizing engineering concepts in sensory and motor rehabilitation. Engineering concept in communication disorders. Rehabs for locomotion, visual, speech & hearing. Artificial limb and hands, prosthetic heart valves. Externally powered and controlled orthotics and prosthetics. Myoelectric hand and arm prostheses. The marcus intelligent hand prostheses, gait study, spinal rehabilitation

TEXT BOOKS:

1. Robbinson C.J., Rehabilitation Engineering. CRC press 1995
2. Gerald E. Miller, Artificial Organs, Morgan & Claypool Publishers, 2006

REFERENCE BOOKS:

1. Bronzino Joseph, Hand book of biomedical engineering. CRC; 2 Sub edition, 1999
2. R.S. Khandpur, Hand book of biomedical instrumentation. Tata Mcgraw Hill Publishers, 1/e.
3. David O. Cooney., Biomedical Engineering Principles (Volume – II). Marcel Dekker Inc.
4. Ballabio E.et al, Rehabilitation Engineering. IOS press 1993.

RBM073 THERAPEUTIC EQUIPMENTS

UNIT I

Cardiac Pacemakers & Defibrillators: Effects of electric field on cardiac muscles and laws of stimulation. External, internal, and Programmable pacemakers. Pulse generator: sensing, output and timing circuits. Power sources, electrodes and leads system, pacing system analyzers. Defibrillators - basic principle and comparison of output wave forms of different DC defibrillator, energy requirements, synchronous operation, implantable defibrillators, defibrillator safety and analyzers, RF ablation treatment for arrhythmia.

UNIT II

Ventilators & Anaesthetic system: Basic principles of ventilators, different generators, inspiratory phase and expiratory phase, different ventillatory adjuncts, neonatal ventilators, p based ventilator, ventilator testing. Anaesthesia: Need of anaesthesia, gas used and their sources, gas blending and vaporizers, anaesthesia delivery system, breathing circuits.

UNIT III

Physical therapy: Physical therapy principles
Electrical stimulators: Strength-duration curve, types of stimulators, an electrodiagnostic / therapeutic stimulator. Nerve-muscle stimulator: peripheral nerve stimulator, Ultrasonic stimulators, stimulators for pain and relief.
Diathermy: IR diathermy, UV diathermy, short wave diathermy, microwave diathermy, ultrasonic diathermy.

UNIT IV

Surgical Diathermy & LASER: Principles and applications of surgical diathermy, Physics and engineering of ultrasonic lithotripter, basic principle of extracorporeal shock wave lithotripter. Principle operation of LASER, various application of CO_2, argon, He - Ne, Nd – YAG & pulsed ruby LASER, Application of LASER in surgery.

UNIT V

Electro-surgery & Neonatal care unit: Electrosurgery machine, electrosurgery circuits, solid state electrosurgery generator circuits, electrosurgery safety, testing electrosurgery units, cautery, light sources, suction apparatus, and sterilizers. Baby incubator, radiant warmer and phototherapy unit.

TEXT BOOKS:

1. R. S. Khandpur “Handbook of Bio-Medical Instrumentation”, Tata McGraw Hill.
2. Carr & Brown, “Introduction to Biomedical Equipment Technology” Pearson Education, Asia.
   J.Webster, “Bioinstrumentation”, Wiley & Sons

REFERENCES:

1. Joseph Bronzino, “ Biomedical Engineering and Instrumentation”, PWS Engg . , Boston. Willard Van Nostrand, “.Instrumental Methods of Analysis” Sharms, “Instrumental Methods”, S Chand & Co.
2. Harry Bronzino E, “Handbook of Biomedical Engineering and Measurements”, Reston, Virginia. Jacobson & Websler, “ Medicine & Clinical Engg”
3. Leslie Cromwell, “ Biomedical Instrumentation and Measurements” Hcinz Kressc, “ Handbook of Electro medicine” , John Wiley.
4. Geddes & Baker , “Principles of Applied Biomedical Instrumentation” Wiley.

DEPARTMENTAL ELECTIVE IV

RBM074 TELEMEDICINE

UNIT I

Fundamental concepts – Significance, Principle, functional blocks of Telemetry and Telecontrol system-Methods of telemetry – Electrical, Pneumatic, Hydraulic and Optical Telemetry – State of the art-Telemetry standards.

UNIT II

Clinical network, Clinical parameters, Cardiology, Dermatology, Tele-radiology, EMI emergency medicine, Gastroenterology, Homecare, Neurology, Oncology, Ophthalmology, Mental health, Tele-rehabilitation, Tele-pathology & Tele-surgery.

UNIT III

Use of computers in distance mode of healthcare delivery, Web technology, Satellite communication systems; hypertext, voice & image transfer protocols, Medical image scanning, Data compression and Transfer, Capturing of medical signals, Analog to digital conversion, Video conferencing, Remote sensing, Rural primary setups, Referral and Super speciality centers, Societal medico legal aspects, Networking (local, national & global).

UNIT IV

Video conferencing hardware/software, Video production, Editing and Broadcasting, Tele-medical workstations, DSL equipments, Cable modem, POTS line, Fast switches ethernet, Fiber optic equipment, Router, Hubs, Monitoring devices, Vital sign monitoring devices, Respiratory monitoring devices, Neurological monitoring devices, Video scopes, Robotics and virtual reality devices

UNIT V

Legal and ethical issues, Duty of care, Malpractice and liability, Licensure and accreditation, Security and confidentiality, Ethical standards, Intellectual property rights

TEXT BOOKS:

1. B.D. Gupta, “Introducing Telemedicine (Applications, challenges, needs and benefits, components and infrastructure)”

2. A.C. Norris, “Essentials of Telemedicine and Telecare”

3. Marlene Maheu, Pamela Whitten, Ace Allen, “E-health, Telehealth and Telemedicine”

4. Marilyn J. Field, Telemedicine: A Guide to Assessing Telecommunications for Health Care, National Academic Press, 1996

REFERENCE BOOKS

1. Charles J. Amlaner (Author), David W. Macdonald (Author), A Handbook on Biotelemetry and Radio Tracking, Pergamon Press; 1st edition (January 1, 1980)

RBM075 BIOTRANSPORT PHENOMENA

UNIT 1

Introduction to fluid mechanics, heat and mass transfer. Physical, chemical and rheological properties of blood.

UNIT II

Unified approach of momentum, heat and mass transfer. Heat Transport: Heat production in humans, Loss of heat to the environment, Heat transfer within the body

UNIT III

Transport through cell membranes: Membrane structure, composition and permeability, Osmosis, Passive diffusion, Pressure diffusion, Facilitated transport, Facilitated diffusion of oxygen in haemoglobin solutions, Active transport, Pinocytosis.

UNIT IV

Compartment modeling: Pharmacokinetic models, The one-compartment and two-compartment open models. Structure and gross operational features of the respiratory system, Gas transport mechanisms in the lungs, Oxygen and carbon dioxide transfer in the blood, Modeling oxygen uptake in the pulmonary capillaries.

UNIT V

Structure and general features of operation of kidneys, Transport mechanisms in the tubules, Pore models of the glomerular tuft, Countercurrent mechanism of urine formation, Models of nephron function, Analytical model for Henle's loop. Artificial kidney devices: Hemodialysis, types of hemodialyzers.

TEXT BOOKS:

1. David O. Cooney, An introduction to fluid, heat & mass transport process- Principles, Vol.1, Marcel Dekker Inc., New York, 1976.
2. EdwinN. Lightfoot, Transport phenomena and living systems – Biomedical aspects of momentum and mass transport, John Wiley, 1974
3. Ronald L. Fournier, Basic transport phenomena in biomedical engineering, Taylor Francis, 1998.

RBM076 DATA COMMUNICATION NETWORKS

UNIT I

Introduction to Networks and Data Communications, Goals and Applications of Networks, Network structure and architecture, The Internet, Protocols and Standards, Layered Tasks, The OSI reference model, TCP / IP, Addressing, Line Coding Review.

UNIT II

Physical Layer, Transmission Media: Guided and unguided, Network Topology Design, Data Link Layer: Error detection and Correction, Framing, Flow and Error Control Protocols, Networking devices.

UNIT III

Multiple Access: Random Access Protocols, CDMA, CSMA/CD, CSMA/CA, Controlled Access, Channelization Wired LANs: IEEE Standards, Fast Ethernet, Gigabit Ethernet, Wireless LAN IEEE 802.11, Bluetooth IEEE 802.16

UNIT IV

Multiple Access: Random Access Protocols, CDMA, CSMA/CD, CSMA/CA, Controlled Access, Channelization Wired LANs: IEEE Standards, Fast Ethernet, Gigabit Ethernet, Wireless LAN IEEE 802.11, Bluetooth IEEE 802.16

UNIT V

Application Layer: File Transfer, Electronic mail, Virtual Terminals, Cryptography, Network Security

TEXT BOOK:

1. Forouzan, Data Communication & Networking, McGrawhill Education
2. Lathi, B. P. & Ding, Z., (2010), Modern Digital and Analog Communication Systems,Oxford University Press
3. Stallings, W., (2010), Data and Computer Communications, Pearson.
4. Andrew S. Tanenbaum, “Computer Networks” Pearson.
5. Ajit Pal, “Data Communication and Computer Networks”, PHI
6. Dimitri Bertsekas, Robert G. Gallager, “Data Networks”, Prentice Hall, 1992

SEMESTER – VIII

DEPARTMENT ELECTIVE V

RBM081 ADVANCED BIOMEDICAL INSTRUMENTATION

UNIT I

Analytical equipments: Colorimeter-principles of measurement and applications, Beer-Lambert's Law in spectrometry. UV, visible and infra-red spectrophotometers. Design of monochromators, detection systems. Basic applications in Biochemical analysis-Autoanalyser. Principles and applications - atomic absorption photometer, flame photometers, densitometers, gas and liquid chromatographs. Principles of scanning and transmission electron microscopy. Principles of simple, compound and phase contrast microscopes. Centrifuge-principles and applications. Different types of sterilization methods-autoclave. Blood cell counters: Different methods for cell counting, Coulter Counters, automatic recognition and differential counting of cells.

UNIT II

Blood Flow meters: Electromagnetic blood flow meter, ultrasonic blood flow meter, Doppler blood flow meter, NMR blood flow meter, cardiac output measurement – indicator dilution methods and impedance technique.

UNIT III

Pulmonary function analyzers: Pulmonary function measurement-spirometry, respiratory gas analyzers, pneumotachography – different types of pneumotachometers, respiratory rate meter, impedance plethysmograph / pneumograph.Blood gas analyzers: Blood pH measurement, pCO2 measurement, pO2 measurement, a complete blood gas analyzer. Different types of oximetry systems, pulse oximeter.

UNIT IV

Blood pressure and heart sound measurement: Measurement of blood pressure using sphygmomanometer instrument based on Korotkoff sound, indirect measurement of blood pressure, automated indirect measurement, and specific direct measurement techniques. Heart sound measurement – stethoscope, phonocardiograph.

UNIT V

Endoscopy: Introduction, various types of endoscopes, cystoscopes, laproscopes, fiber optic endoscopes and endoscopes with integral TV cameras.

TEXT BOOKS:

1. R. S. Khandpur “Handbook of Bio-Medical Instrumentation”, Tata McGraw Hill.
2. Carr & Brown, “Introduction to Biomedical Equipment Technology” Pearson Education, Asia.
3. J.Webster, “Bioinstrumentation”, Wiley & Sons

REFERENCES:

1. Joseph Bronzino, “ Biomedical Engineering and Instrumentation”, PWS Engg . , Boston. Willard Van Nostrand, “.Instrumental Methods of Analysis”-Sharms, “Instrumental Methods”, S Chand & Co.
2. Harry Bronzino E, “Handbook of Biomedical Engineering and Measurements”, Reston, Virginia.
3. Jacobson & Websler, “ Medicine & Clinical Engg”
4. Leslie Cromwell, “ Biomedical Instrumentation and Measurements”

RBM082 TISSUE ENGINEERING

UNIT I

Introduction: Basic definition, Structural and organization of tissues: Epithelial, connective; vascularity and angiogenesis, basic wound healing, cell migration, current scope of development and use in therapeutic and in-vitro testing.

UNIT II

Cell culture: Different cell types, progenitor cells and cell differentiations, different kind of matrix, cell-cell interaction. Aspect of cell culture: cell expansion, cell transfer, cell storage and cell characterization, Bioreactors.

UNIT III

Molecular biology aspects: Cell signaling molecules, growth factors, hormone and growth factor signaling, growth factor delivery in tissue engineering, cell attachment: differential cell adhesion, receptor-ligand binding, and Cell surface markers.

UNIT IV

Scaffold and transplant: Engineering biomaterials for tissue engineering, Degradable materials (collagen, silk and polylactic acid), porosity, mechanical strength, 3-D architecture and cell incorporation. Engineering tissues for replacing bone, cartilage, tendons, ligaments, skin and liver. Basic transplant immunology, stems cells: introduction, hepatopoiesis.

UNIT V

Case study and regulatory issues: Case study of multiple approaches: cell transplantation for liver, musculoskeletal, cardiovascular, neural, visceral tissue engineering. Ethical, FDA and regulatory issues of tissue engineering.

TEXT BOOK:

1. Clemens van Blitterswijk, Tissue Engineering, Academic Press, 2008

REFERENCE BOOKS:

1. Principles of tissue engineering, Robert. P.Lanza, Robert Langer & William L. Chick, Academic press.
2. The Biomedical Engineering –Handbook, Joseph D. Bronzino, CRC press.
3. Introduction to Biomedical Engg. , Endarle, Blanchard & Bronzino, Academic press.
4. Tissue Engineering, B. Palsson, J.A. Hubbell, R.Plonsey & J.D. Bronzino, CRC- Taylor & Francis

RBM083 PRINCIPLES OF RADIO DIAGNOSIS AND RADIO THERAPY

UNIT - I

Production of X-rays Various components of radiographic systems Electrical circuit for X-ray unit filament circuits and mA control- IITV circuits - KV control exposure switching and control of exposure timers - types of X -ray tubes for various medical application. Rating charts of X -ray tubes.

UNIT - II

Scattered radiation & its control in radiography collimators bucky grids absorbed dose - Basics of tables & arms. Fluoroscopy systems TV chain for fluroscopy Properties of X-ray films &screens- Characteristics of imaging system - modulation transfer function.

UNIT - III

Automatic exposure controls - Photo timers - types - limitations - performance - serial film chargers types - radiographic considerations - film exposure time - photo timer applications - automatic brightness control system.

UNIT - IV

Basic of digital angiography - Image processors for digital angiography - processor architecture -Temporal integration techniques for digital angiography- digital subtraction angiography

UNIT - V

Physical principles of radio therapy. Dosage data for clinical applications. Measurement of output and use of ISODOSE charts. Collimators and beam direction devices. Telemetry sources and acceptancecalibration. Safety protocols & protection. Principles of linear accelerators for radiation therapy.Radiation therapy planning.

References:

Massey & Meredith , "Fundamental Physics of Radiology", John Wright & SonsWebb S, " The Physics of Medical Imaging", Adam Hilger, Bristol Thomas Thompson , "A Practical Approach to Modern Imaging Equipment" ,.Little Brown cloSybil M Stockley, " A Manual of Radiographic Equipments", Churchill Livingstones Chistrmis, "Physics of Diagnostic Radiology"

DEPARTMENTAL ELECTIVE VI

RBM084 DESIGN AND MODELLING OF BIOMEDICAL SYSTEMS

UNIT-I SELECTION OF HARDWARE

Sensor selection and their specifications, Power Supply Design, Filter Design, Amplifier Design, Display Devices, Data Recorders.

UNIT-II

Simulation of Design on using Simulation Softwre.

UNIT-III PCB DESIGN AND FABRICATION TECHNIQUES

PCB Design Software, types of PCB Design, Different Method of PCB Making.

UNIT-IV RELIABILITY ASPECT OF BIOMEDICAL EQUIPMENTS

Basic Reliability related Definitions, Need of reliability in Medical Device, Medical device reliability and Associated areas, Basic reliability Mathematics and Concepts for Medical Devices, Reliability Configuration.

UNIT-V SFETY ASPECT OF BIOMEDICAL EQUIPMENTS & THEIR CALIBRATION AND TESTING

Electrical shock hazards, Leakage current, proper Ground, Shielding, Safety codes for electro medical Equipments, Electrical safety Analyzer.

Calibration and Testing of Equipments.

REFERENCE BOOKS:

1. Balbir S Dhillon, "Medical Device Reliability and Associated Areas", Pub-CRC Press.
2. Khandpur R.S., "Hand book of Biomedical Instrumentation", Tata McGraw Hill, 2004.

RBM085 COMMUNICATION ENGINEERING

UNIT-I

Need for modulation - Amplitude modulation – Frequency spectrum of AM wave – Representation of AM – Power relation –Frequency modulation – Frequency spectrum of FM wave –AM transmitter – FM transmitter – Super heterodyne AM receiver –FM receivers.

UNIT-II

Principles of pulse modulation – sampling theorem, PAM – PWM – PPM– Conversion of PWM wave to PPM wave – Generation of PAM, PPM and PWM waves – Demodulation of PAM, PWM, PPM – An introduction to digital modulation systems – PCM, ASK, FSK and PSK.

UNIT- III

Microwave communication systems: advantage, block diagram of a microwave radio system, microwave radio stations- Terminal station and repeater station.

Satellite Communication system: Satellite Orbits, launch vehicles, look angles, satellite parameters, satellite link model, personal communication systems- GPS services.

UNIT- IV

Amount of information, Entropy, Information rate, Shannon's theorem, Channel capacity, Bandwidth and S/N trade off, Introduction to error and error correction code.

UNIT -V

Cellular concept, basic cellular concept and its operation, uniqueness of mobile radio environment- Performance metrics in cellular system-Elements of cellular mobile radio-Handoff-Frequency management and channel assignment- Introduction to various cellular standards like AMPS, GSM, GPRS, IS-95A, IS-95B, CDMA-2000 and WCDMA.

TEXT BOOK:

1. Kennedy Davis, "Electronic Communication Systems", Tata McGraw Hill Publishing Company Limited, New Delhi, 1999.
2. Wayne Tomasi, "Electronic Communication Systems", Pearson education Private Limited, Delhi, 2004.

REFERENCE BOOKS:

1. Roddy D and Coolen J, "Electronic Communications", Prentice Hall of India Private Limited, fourth edition, 2007.
2. William C.Y. Lee, "Mobile Cellular Telecommunication Systems", McGraw Hill International

Gerd Keiser, "Optical fiber Communications", McGraw Hill International Edition, Fourth edition, 2006.

RBM086 Artificial Intelligence & its Applications in Biomedical Engineering

UNIT-I

Introduction to Artificial neuron and neural networks, Feature selection. Types of learning, supervised and unsupervised learning, Supervised learning decision surfaces, Two, category separation, linearly separable sets, Multiple category classification problems, Relationship to neural network models, Comparison of methods, Applications.

UNIT-II

Unsupervised learning, Clustering, Kohonen network and competitive learning. Hebbian learning, Adaptive resonance theory (ART), Applications.

UNIT-III

Introduction, Foundation of Fuzzy system, Fuzzy systems at work; Fuzzy system design, Crisp V/s Fuzzy sets, Fuzzy sets to fuzzy event, Fuzzy logic, Practical fuzzy measures, Fuzzy set operations, properties of fuzzy sets, Fuzzification techniques, Relational inference, Compositional inference. Linguistic variables and logic operators, Inference using fuzzy variables, Fuzzy implication.

UNIT-IV

Fuzzy systems and algorithms, Defuzzification, Adaptive fuzzy system algorithms, Expert systems v/s fuzzy inference engines, Basic fuzzy inference algorithm, Overall algorithm, Input data processing, Evaluating antecedent fuzzy variables, Left hand side computations; Right hand side computations, Output processing.

UNIT-V

Introduction to Genetic Algorithm, Application of AI in biomedical engineering.

TEXT BOOKS:

1. Donna L. Hudson and Maurice E. Cohen., "Neural Networks and Artificial Intelligence for Biomedical Engineering", Prentice Hall of India. Pvt. Ltd., New Delhi
2. Riza C. Berkan and Sheldon L. Trubatch., "Fuzzy systems Design Principles", Standard Publishers and Distributors, Delhi.
3. Abraham Kandel and Gideon Langholz, "Fuzzy Control Systems", CRC Press, Boca Raton.
4. J.S.R. Jang, C.T. Sun and E. Mizutani, "Neuro, Fuzzy and soft computing", Prentice Hall of India. Pvt. Ltd., New Delhi.